

# WASP-2b

R-0065

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-2\_211020 · created 2026-07-15T15:44:26+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459507.6663 +/- 0.0059 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.16 +/- 0.012                  |
| Transit depth (Rp/Rs)^2                           | 2.56 +/- 0.39 [%]               |
| Orbital Inclination (inc)                         | 85.39 +/- 1.58                  |
| Airmass coefficient 1 (a1)                        | 1.002 +/- 0.017                 |
| Airmass coefficient 2 (a2)                        | -0.002 +/- 0.012                |
| Scatter in the residuals of the lightcurve fit is | 1.69 %                          |
| Best Comparison Star                              | #3 - [244, 98]                  |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.078 +/- 0.016                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.16 $\pm$ 0.012 vs 0.1326 (deviation 2.28 $\sigma$ ) |
| Duration fit vs published            | 0.078 d vs 0.0737 d (deviation 0.27 $\sigma$ )        |
| Mid-time O–C                         | 8.7 min                                               |
| Depth SNR                            | 13.3                                                  |
| Residual scatter                     | 1.69 %                                                |

### Light curve

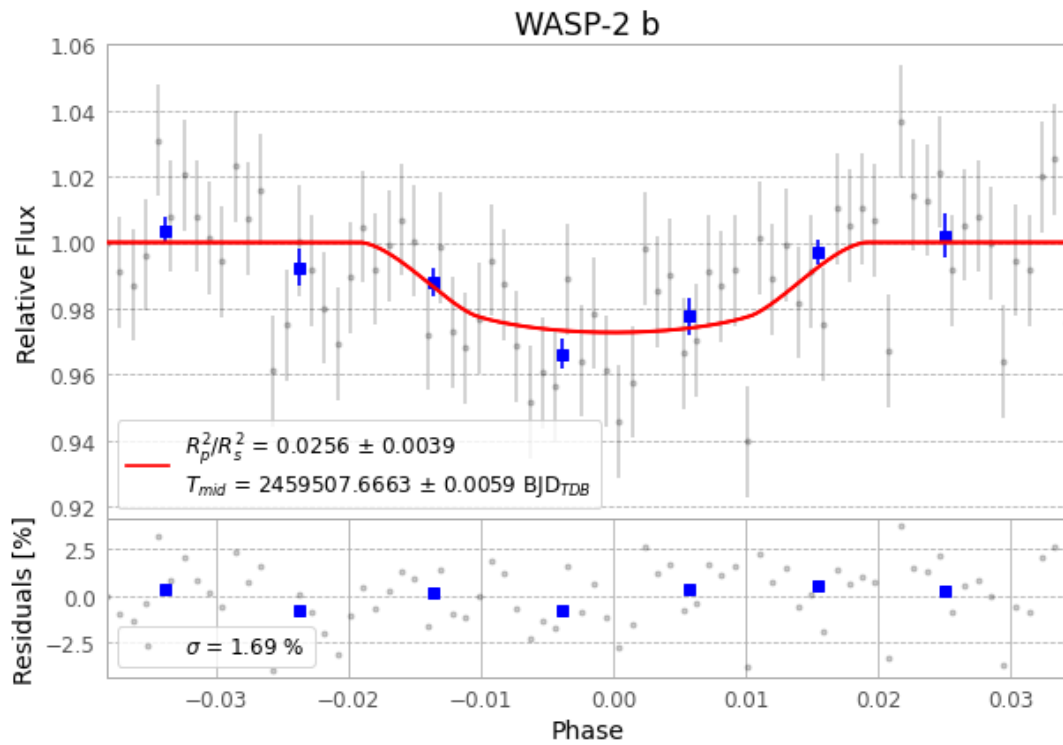


Figure 1 — FinalLightCurve\_WASP-2 b\_2021-10-19.png (SHA-256 496a273ccd2b8b44...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 123], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 52d5d17133063e68...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

## Data provenance

76 FITS frames (50 MB), WASP-2211020015712.FITS → WASP-2211020054212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-211020.log (c8712eb6b601...), FinalLightCurve\_WASP-2 b\_2021-10-19.png (496a273ccd2b...), FinalLightCurve\_WASP-2 b\_2021-10-19.csv (f48332778b44...)

## Compute resources

Wall time 140.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.